Acid-enhanced ternary DES pretreatment of wheat straw guided by Kamlet-Taft parameters: Boosting lignin separation and enzymatic hydrolysis

Abstract: In this study, ternary deep eutectic solvent (DES) systems were constructed by introducing solid acids ($AlCl_3$, PTA, $FeBr_3$) into binary DES systems to achieve improved lignocellulose separation efficiency and enhanced enzymatic hydrolysis. Correlation analysis between the Kamlet-Taft solvatochromic parameters and lignin extraction rate/enzymatic hydrolysis efficiency revealed that the pretreatment and enzymatic hydrolysis steps were significantly improved by enhancing the hydrogen bond acidity. Accordingly, ternary DES systems with better pretreatment efficacy were screened. Notably, using the ChCl/BDO/$AlCl_3$ (25:50:1) system at 120 °C for 60 min provided the most optimal pretreatment and enzymatic hydrolysis performance for wheat straw. After pretreatment under these conditions, an 85.74% lignin extraction rate was achieved, and 72 h of enzymatic hydrolysis provided a 99.67% polysaccharide degradation rate and 74.59% reducing sugar yield. Moreover, this pretreatment strategy demonstrated good applicability to multiple biomass types, providing a novel approach for developing efficient and environmentally friendly lignocellulose separation technologies.

Keywords: Lignocellulosic biomass; Deep eutectic solvents; Enzymatic hydrolysis; Lignin; Pretreatment

1. Introduction

Given the unsustainability of non-renewable resources and persistent ecological degradation, lignocellulosic biomass is being increasingly recognized as a promising green, renewable, and

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sustainable resource (Yiin et al., 2021). Lignocellulosic biomass is typically composed of polysaccharides (cellulose and hemicellulose) and lignin, which is an aromatic polymer (Baraka et al., 2025). However, the dense structure, complex composition, cellulose crystalline regions, and lignin hydrophobic groups in lignocellulosic biomass promote the formation of stable covalent bonds and hydrogen bond networks between biopolymers. Consequently, the structure of lignocellulosic biomass is resistant to disruption, hindering its efficient utilization (Baraka et al., 2025). To enhance the utilization efficiency of lignocellulosic biomass, a pretreatment step is required to alter the native structure and separate the target components (Mankar et al., 2021).

Deep eutectic solvents (DESs) are an emerging class of green solvents that combine the advantages of ionic liquids and organic solvents. DES systems are widely applied for the clean and efficient separation of lignocellulosic components due to their simple synthesis, low cost, strong stability, good biocompatibility, and recyclability (Li et al., 2024). When utilized as pretreatment agents, DES systems form strong hydrogen bonds with the components of lignocellulosic biomass, competitively disrupting the original hydrogen bond network to effectively separate the individual components (Chen et al., 2018). DESs are formed by mixing specific stoichiometric ratios of hydrogen bond acceptors (HBA) and hydrogen bond donors (HBD) (e.g., amides, carboxylic acids, polyols) (Sunar et al., 2024). Choline chloride (ChCl) is widely utilized as an HBA for lignin separation due to its economic viability and biocompatibility (Sunar et al., 2024). For instance, Alvarez et al. (Alvarez-Vasco et al., 2016) reported that using a lactic acid/choline chloride DES to treat poplar at 145 °C for 6 h resulted in a lignin yield of 78.5%. Tan et al. (Tan et al., 2018) compared the lignin separation efficiencies achieved by using different DESs to treat oil palm empty fruit bunches at 120 °C for 8 h. They showed that the lignin separation performance of acidic DESs was significantly better than that

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of neutral and alkaline DESs. Among them, acidic choline chloride/lactic acid exhibited optimal component separation performance, with a 100% hemicellulose extraction rate, 88% lignin removal rate, and 50% lignin extraction rate. These findings provide critical solvent selection criteria for efficient lignin separation. In a study on multicomponent DESs, Jiang et al. reported that introducing an acidic third component into binary DES systems markedly enhanced their lignocellulose separation efficiency (Jiang et al., 2020b). The improved performance of ternary DES systems is partly because the high viscosity of DESs adversely affects biomass delignification (Mankar et al., 2021). Introducing a third component can reduce the viscosity of DESs while enhancing their solubility and stability (Arshad et al., 2025). Moreover, the addition of a third component also provides active protons and additional acidic sites as well as additional functionality (Jiang et al., 2020a). The incorporation of metal chlorides enables the selective catalytic degradation of hemicellulose. Adding such metal cations to DES systems increases the number of acidic active sites, leading to improved biomass component separation (Feng et al., 2024; Wang et al., 2020).

DES solvent systems demonstrate favorable extraction performance in lignin separation applications, achieving lignin dissolution rates that generally exceed 50%. However, the industrial implementation of these systems still faces challenges. For instance, existing processes commonly require high operating temperatures and prolonged reaction times, leading to significantly higher production costs and impeding the development of continuous production processes (Rajapakse et al., 2025). Therefore, developing more efficient and milder methods to enhance DES-based lignin separation will be crucial for industrial applications. Building upon binary DES systems for wheat straw lignin separation, this study constructed ternary DES systems by introducing solid acids as catalysts. These ternary DESs were used to pretreat the wheat straw with the goal of achieving an

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improved lignin extraction rate and enzymatic hydrolysis efficiency. The ternary DES systems that demonstrated solvatochromic parameters performance were screened by measuring their solvatochromic parameters for DES performance optimization. Then, the reaction parameters were optimized, and the identified optimal conditions were applied to pretreat other biomass types, providing insights for green and efficient lignin separation.

2. Materials and Methods

2.1. Materials

Wheat straw, rice straw, corncob, and corn straw were sourced from the Wheat Experimental Base of Henan Agricultural University. Eucalyptus wood was provided by the College of Forestry, Henan Agricultural University. All natural materials were sun-dried, pulverized, dewaxed, sieved through a 60-mesh sieve, dried, and stored in desiccators for later use. Choline chloride (ChCl), 1,4-butanediol (BDO), ethylene glycol (EG), glycerol (GL), aluminium chloride ($AlCl_3$) and ferric bromide ($FeBr_3$) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. PTA was obtained from Tianjin Damao Chemical Reagent Factory. Other reagents were of analytical grade.

2.2. Preparation of DESs

The reagents ChCl/BDO (1:2), ChCl/EG (1:2), ChCl/GL (1:2, 1:4), ChCl/BDO/$AlCl_3$ (25:25:0.5, 25:50:0.5, 25:50:1), ChCl/EG/PTA (25:25:0.025, 25:50:0.025, 25:50:0.05), and ChCl/GL/$FeBr_3$ (25:50:0.25, 25:100:0.25, 25:100:0.5) were accurately weighed according to the required molar ratios. Each mixture was stirred at 70 ℃ until a homogeneous transparent liquid was formed to obtain the corresponding DES. The resulting DESs were placed in desiccators for further use.

2.3. Determination of Kamlet-Taft solvatochromic parameters of DESs

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Nile red (Jiang et al.), 4-nitroaniline (NH_2), and N,N-diethyl-4-nitroaniline (NEt_2) were dissolved in anhydrous methanol to prepare 0.05 mM dye solutions, which were protected from light before their further use. 5 mL aliquots of the dye solutions were transferred to 10 mL centrifuge tubes, and the tubes were vacuum-dried to remove the methanol. Then, 3 mL of a different DES was added to each tube and thoroughly mixed with the dyes. Detection was performed using a UV-vis spectrophotometer within the scanning wavelength range of 300–700 nm. The following equations were used to calculate the Kamlet-Taft solvatochromic parameters α, β, and π* of the different DESs (Xia et al., 2018):

$$\pi^* = 0.314 \times (27.52 - \nu(NEt_2))\#(1)$$

$$\alpha = \frac{19.9657 - 1.0241 \times \pi^* - \nu(NR)}{1.6078}\#(2)$$

$$\beta = \frac{11.134 - 3580}{\lambda_{max}(NH_2)} - 1.125 \times \pi^*\#(3)$$

Where: α is the hydrogen bonding acidity; β is the hydrogen bonding basicity; π* is the polarizability; λ_{max} is the maximum absorption wavelength of each dye; ν is the staining wavenumber of each dye, $\nu=1/(\lambda_{max}\times10^{-4})$.

2.4. Pretreatment of wheat straw with DESs

Wheat straw was uniformly mixed with a DES using a solid-to-liquid ratio of 1:20. Each mixture was heated under stirring at a specified temperature (100 ℃, 110 ℃, 120 ℃, 130 ℃, 140 ℃) for a defined duration (20 min, 30 min, 40 min, 60 min, 80 min, 100 min). An anhydrous ethanol-water solution (7:3 v/v) was added to the pretreated mixtures, which were then magnetically stirred for 30 min. Solid-liquid separation was achieved through repeated centrifugation and washing. The supernatant from each mixture was rotary-evaporated to remove the ethanol, yielding a lignin-DES solution. Deionized water was then added to this lignin-DES solution, and the combined solution was

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stored at 4 ℃ for static precipitation. Finally, lignin was separated by centrifuging the mixture, and freeze-drying was employed to obtain an absolute dry lignin sample.

2.5. Composition analysis

The polysaccharide and lignin contents of the untreated and pretreated wheat straw were determined according to NREL analytical procedures (Hou et al., 2018). Specifically, 30 mg samples of wheat straw or pretreated residues were hydrolyzed with 0.3 mL of 72% sulfuric acid at 30 °C for 1 h. Each mixture was then diluted to 4% acid concentration by adding 8.4 mL of distilled water, and hydrolysis was performed at 121 °C (0.1 MPa) for 1 h. After the reaction, the hydrolysates were filtered through pre-dried (105 °C to constant weight) G4 glass sintered funnels under a vacuum. The filtrates were then divided: one portion was neutralized with calcium carbonate to determine the reducing sugar concentration via the 3,5-dinitrosalicylic acid (DNS) method, enabling the polysaccharide content to be calculated. The other portion was analyzed via UV spectrophotometry at 320 nm, and the acid-soluble lignin (ASL) content was calculated using an extinction coefficient of 30 $L \cdot g^{-1} \cdot cm^{-1}$. The residual solids adhered to the tube were rinsed into funnels with distilled water and repeatedly washed to prevent the carbonization of the carbohydrates during drying. The residues were then oven-dried at 105 °C for ≥4 h to a constant weight and weighed. Subsequently, the residues were ashed in a muffle furnace at 575 °C for 3 h to a constant weight to quantify the acid-insoluble lignin (AIL) content. The lignin extraction rate and residue recovery rate were calculated using the following formulas:

$$Lignin\ extraction\ rate(\%) = (1 - \frac{Lignin\ amount\ in\ the\ recovered\ residue}{Lignin\ amount\ in\ the\ native\ biomass}) \times 100\#(4)$$

$$Residue\ recovery\ rate(\%) = \frac{Recycled\ residue\ amount}{Native\ biomass\ amount} \times 100\#(5)$$

2.6. Enzymatic hydrolysis analysis

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20 mg of untreated or pretreated wheat straw, cellulase (40 FPU/g), 7 mL preheated (50 °C) citrate buffer (50 mM, pH 4.8), and 20 μg/mL sodium azide were weighed and added to a 25 mL ground-glass stoppered conical flask. The timed reaction was performed in a constant-temperature incubator shaker at 50 °C with 200 r/min agitation. At regular intervals, 0.3 mL aliquots were removed and inactivated in a boiling water bath for 5 min to terminate the enzymatic reactions. Centrifugation (10,000 r/min, 5 min) was then performed using a high-speed refrigerated centrifuge, and the DNS method was used to determine the reducing sugar concentration. The polysaccharide degradation rate and reducing sugar yield were calculated using the following formulas:

$$Polysaccharide\ degradation\ rate(\%) = \frac{Rleased\ reducing\ sugar\ amount\ in\ enzymatic \times 0.9}{Polyaccharide\ amount\ in\ the\ \text{enzymatic sample}} \times 100\#(6)$$

$$Reducing\ sugar\ yield(\%) = \frac{Rleased\ \ reducing\ sugar\ amount}{Theoretic\ reducing\ sugar\ amount\ in\ the\ untreated\ samples} \times 100\#(7)$$

Where: 0.9 is the factor used to convert the reducing sugar content to the equivalent polysaccharide content.

2.7. Determination of reducing sugar concentration by the DNS method

The reducing sugars were quantified according to the IUPAC DNS methodology (Soares et al., 2017). First, 0.2 mL of centrifuged enzymatic hydrolysate was diluted to 2 mL with distilled water. Next, 3 mL of DNS reagent was added, and a reaction was performed in a boiling water bath for 5 min. The reacted mixture was cooled to room temperature using cold water and then appropriately diluted with distilled water. Finally, the absorbance was measured at 540 nm using a spectrophotometer.

2.8. Characterization of natural materials and solid residues

Fourier transform infrared spectroscopy (FTIR, Nicolet iS50, Thermo, USA) was utilized to characterize the natural biomass materials and solid residues pretreated with the DESs. Each sample

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was thoroughly ground, mixed with potassium bromide (KBr) powder, and then pressed into a transparent pellet. Full-range scanning (400–4000 cm^{-1}) was performed using an FTIR spectrometer at a resolution of 0.4 cm^{-1}. Structures were analyzed in transmittance mode.

X-ray diffraction (XRD, ULTIMA IV, Rigaku, Japan) was utilized to evaluate the crystal structures of the biomass samples before and after pretreatment. Diffraction patterns were obtained in the 2θ range of 5°–60° using a speed of 5°/min, step size of 4°, voltage of 40 kV, and current of 40 mA. The crystallinity index (CrI) was calculated using the following formula:

$$CrI(\%) = \frac{I_{002} - I_{am}}{I_{002}} \times 100\#(8)$$

Where: I_{002} is the diffraction intensity of the 002 crystal plane (2θ ≈ 22°) and I_{am} is the diffraction intensity in the amorphous region (2θ ≈ 18°).

Scanning electron microscopy (SEM, Gemini 300, Zeiss, Germany) was used to analyze the surface morphology of the pretreated materials. Before testing, the sample powders were uniformly dispersed on metal stubs with a conductive adhesive and then sputter-coated with gold.

2.9. Statistical analysis

All laboratory analyses were performed in triplicate to ensure reproducibility. The figures in this work present results as the mean ± standard deviation (error bars).

3. Results and Discussion

3.1. Analysis of DES solvatochromic parameters

Kamlet-Taft solvatochromic parameters are widely used to characterize the microscopic physicochemical properties of solvents. The parameters α and β respectively represent hydrogen bond acidity (proton-donating ability) and basicity (proton-accepting ability), while π* indicates solvent dipolarity/polarizability (Florindo et al., 2017). The π* value reflects the strength of cation-anion

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interactions. Meanwhile, higher α and β values significantly impact pretreatment efficiency: elevated α values facilitate proton dissociation, while increased β values enhance the acceptance of hydrogen bonds by anions (Weng & Toner, 2018). Thus, the Kamlet-Taft solvatochromic parameters of a DES can quantify the ease of hydrogen bond formation, enabling a preliminary prediction of pretreatment efficiency. As shown in Table 1, adding solid acids ($AlCl_3$, PTA, $FeBr_3$) to the prepared DESs as third components increased their π* and α values to varying degrees. Higher α values indicate enhanced solvent acidity, which may promote ether bond cleavage in lignocellulose. DESs with high α values readily donate hydrogen bonds to release H^+, making α a critical factor for lignin extraction efficiency(Liang et al., 2021). Solvent polarity (indicated by π*) affects activation energy and solubility. The ternary DESs exhibited higher π* values than the binary systems, indicating stronger ion-ion interactions. Based on the enhancement of solvent acidity and polarity reflected by the elevated α and π values observed in Table 1, it was preliminarily predicted that the three ternary DES, ChCl/BDO/$AlCl_3$ (25:50:1), ChCl/EG/PTA (25:50:0.05), and ChCl/GI/$FeBr_3$ (25:100:0.5), had a better pretreatment for wheat straw Effect.

3.2. Effects of DES composition on wheat straw pretreatment and enzymatic hydrolysis

The impact of DES composition on wheat straw pretreatment was evaluated under standardized conditions (120 °C, 1 h), as shown in Fig. 1. The diol-based binary DES systems exhibited uniformly low pretreatment and enzymatic hydrolysis efficiencies, with statistically insignificant variations. However, the incorporation of solid acids ($AlCl_3$, PTA, $FeBr_3$) to obtain ternary DES formulations significantly enhanced both efficiency metrics relative to the binary systems. The ChCl/BDO/$AlCl_3$ (25:50:1) system exhibited the best performance, achieving 85.74% lignin extraction, 99.67% polysaccharide degradation (after 72 h), and a reducing sugar yield of 74.59%. These values

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substantially exceeded those of the ChCl/BDO (1:2) benchmark (13.63%, 56.17%, 45.69%), and the residue recovery decreased to 49.92% versus 73.85% for the binary system. Notably, the lignin extraction efficiency demonstrated composition-dependent variation. Both alcoholic and acidic components enhanced delignification, while acidic constituents promoted lignin separation. This was because the acidic components provided supplementary protons, facilitating the cleavage of lignocellulosic bonds (ether/ester linkages, polysaccharides) (Weng & Toner, 2018; Yang et al., 2022). The deposition of lignin on cellulose surfaces has been reported to critically constrain enzymatic hydrolysis through dual mechanisms (Lu et al., 2018). This deposition creates a physical barrier that limits enzyme accessibility, and the irreversible adsorption of cellulase via hydrophobic interactions reduces the effective enzyme concentration (Yao et al., 2021). In this work, ternary DES pretreatment induced progressive xylan/lignin removal at elevated temperatures, resulting in complete cellulose exposure. This structural modification increased the accessibility of cellulose binding sites while reducing non-productive enzyme adsorption, leading to an enhancement in the enzymatic hydrolysis efficiency.

3.3. Correlation assessment of solvatochromic parameters with lignin extraction rate and enzymatic hydrolysis efficiency

Positive correlations were observed between the lignin extraction rates from the DES-pretreated samples and the Kamlet-Taft parameters α and (α-β), as displayed in Fig. 2(a-b). Changing the DES composition affected the lignin extraction efficiency. Solid acid incorporation substantially enhanced delignification, which was attributed to the strong hydrogen bonding between the solid acids and ChCl facilitating the cleavage of lignocellulosic linkages and increasing lignin dissolution (Chen et al., 2022). Positive correlations were also observed between the polysaccharide degradation rates and reducing

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sugar yields (72 h enzymatic hydrolysis) with parameters α and (α-β), as shown in Fig. 2(c-f). The introduction of solid acids significantly improved the saccharification metrics, which was likely due to hydrogen bond-mediated cleavage in the lignocellulose(Liu et al., 2019). When cellulose undergoes enzymatic hydrolysis, residual lignin impedes the hydrolysis process by restricting the accessibility of the enzymes to carbohydrates. After pretreatment by solid acid-catalyzed DES, wheat straw removes the vast majority of lignin, and the contact area of cellulase enzyme with the bare cellulose increases, thus improving the enzymatic efficiency (Yu et al., 2016). DES with a higher α-value has better hydrogen bond supplying ability and can catalyze the breaking of the ether bond, glycosidic bond and LCC linkage, thus more favorable for the separation and extraction of lignin, and DES with higher hydrogen bonding acidity (α-value) is more efficient in the separation of lignin(Ji et al., 2025; Xia et al., 2018). Higher α values were correlated with greater lignin removal during pretreatment, reducing the residual lignin content in solids and improving enzymatic digestibility. Based on these relationships, three optimal ternary DES systems were identified for subsequent experimentation: ChCl/BDO/$AlCl_3$ (25:50:1), ChCl/EG/PTA (25:50:0.05), and ChCl/GL/$FeBr_3$ (25:100:0.5).

3.4. Compositional analysis of lignin and solid residues

The compositional analysis of the lignin and solid residues derived from the evaluated ternary DES pretreatment systems is presented in Fig. 3. As shown in Fig. 3(a), the ChCl/BDO/$AlCl_3$-extracted wheat straw lignin exhibited the highest purity (>75%). This lignin had an acid-soluble lignin (ASL) content of 75.33% and an acid-insoluble lignin (AIL) content of 1.62%, with both values significantly exceeding those of natural wheat straw lignin. As shown in Fig. 3(b), the DES-pretreated residues exhibited substantially reduced lignin content and enhanced polysaccharide levels (74.11–85.10%) compared to the untreated biomass (baseline: 56.77%). The enrichment of polysaccharides was

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ascribed to the progressive xylan/lignin removal that occurred during DES pretreatment, which facilitated the complete exposure of the cellulose. This structural modification has been reported to increase the accessibility of cellulose binding sites while reducing non-productive cellulase adsorption via lignin reduction, leading to enhanced enzymatic hydrolysis yields (Cheng et al., 2022). These findings confirm that ternary DES pretreatment effectively fractionates the lignocellulosic components, achieving efficient delignification and improved enzymatic saccharification.

3.5. Effects of reaction temperature and duration on wheat straw pretreatment

Wheat straw was homogenized with ChCl/BDO/$AlCl_3$ (25:50:1) to investigate the effects of temperature and duration on pretreatment efficiency. As revealed in Fig. 3, increasing the pretreatment temperature led to an initial increase and then decline in the lignin extraction efficiency and enzymatic hydrolysis efficiency, which followed the order: 120°C > 110°C > 130°C > 140°C. Meanwhile, the residue recovery demonstrated the opposite trend. Elevated temperatures promote lignin dissociation, but excessively high temperatures can exceed the thresholds of component decomposition, weakening the hydrogen bonding network and diminishing the DES delignification capacity (Chen et al., 2024). Therefore, optimizing the temperature enhances lignin extraction (Zhao et al., 2025). At a fixed temperature of 120 °C, extending the duration of pretreatment from 20 to 60 min substantially improved lignin extraction (42.64% → 85.74%), polysaccharide degradation (72.60% → 99.67%), and the reducing sugar yield (58.41% → 74.59%). This demonstrates the significant time dependence of the delignification process, where prolonged pretreatment reduces the residual lignin content to enhance enzymatic accessibility. However, a plateau in performance was observed when the pretreatment time was extended to 80–100 min. Consequently, 60 min represents the economically optimal pretreatment duration. According to these results, the optimal reaction parameters for

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maximizing the DES lignin separation capacity and enzymatic hydrolysis efficiency are a pretreatment temperature of 120 °C and duration of 60 min.

3.6. Pretreatment of diverse biomasses with ChCl/BDO/$AlCl_3$ under optimal conditions

The optimal pretreatment conditions were identified as treatment with ChCl/BDO/$AlCl_3$ (25:50:1) at 120 °C for 60 min. Under these parameters, five biomass types (wheat straw, rice straw, corncob, corn straw, eucalyptus wood) were treated with ChCl/BDO/$AlCl_3$, as presented in Fig. 5. This ternary DES system demonstrated high efficacy for the pretreatment of wheat, rice, corncob, and corn straw, achieving lignin extraction rates exceeding 70%. Notably superior performance was achieved for the pretreatment of rice straw, with a lignin extraction rate of 87.98%, polysaccharide degradation rate of 99.81%, and reducing sugar yield of 75.01%. These values all exceeded the metrics for wheat straw. Overall, these results demonstrate broad applicability across diverse lignocellulosic feedstocks.

3.7. Characterization and analysis of raw materials and solid residues

FTIR spectra of the raw wheat straw biomass and solid residues obtained after pretreatment are shown in Fig. 6(a). Compared to the untreated wheat straw, DES pretreatment minimally altered the cellulose structure but significantly modified the lignin and hemicellulose. In the FTIR spectra, the peak at 3402 cm^{-1} was ascribed to lignin hydroxyl stretching vibrations. Cellulose characteristic vibrations were observed at 1047 cm^{-1} (C-O-C glycosidic bonds) and 2886 cm^{-1} (C-H stretching vibrations), while the extent of C-O-C aryl ether linkage cleavage in the lignin was indicated by the peak at 1245 cm^{-1}. Peaks at 1047 cm^{-1} and 1368 cm^{-1} were attributed to cellulose C-O-C deformation vibrations, reflecting pretreatment-induced changes in the crystalline regions. The 1603 cm^{-1} and 1476 cm^{-1} peaks signified aromatic skeletal vibrations from lignin benzene rings, suggesting the potential formation of cellulose derivatives. These aromatic peaks were significantly attenuated after

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pretreatment, and at the same time, the signal intensity at 893 cm^{-1} (β-glycosidic C-H stretching vibrations) was enhanced, corresponding to polysaccharide enrichment. This spectral evolution confirms that ternary DES pretreatment effectively delignified the wheat straw biomass. Elevated peak intensities relative to the raw biomass further demonstrate that hydrogen-bond reorganization occurred within the cellulose (Liu et al., 2023).

XRD analysis of the wheat straw biomass samples before and after pretreatment is shown in Fig. 6(b). The crystallinity index (CrI) of the untreated biomass was 33.68%, while pretreatment with ChCl/BDO/$AlCl_3$, ChCl/EG/PTA, and ChCl/GL/$FeBr_3$ led to CrI values of 43.78%, 37.64%, and 35.81%, respectively. This improvement in crystallinity was due to the removal of amorphous lignin and hemicellulose during pretreatment. Although ternary DES pretreatment affected the diffraction peak intensities, the cellulose crystalline structures remained unaltered after pretreatment, which was consistent with the FTIR analysis. Notably, ChCl/BDO/$AlCl_3$ pretreatment led to the most significant increase in CrI, which was due to the more effective removal of amorphous components by this system leading to higher crystallinity (Loow et al., 2018).

SEM images of the untreated and pretreated wheat straw are displayed in Fig. S1. The untreated wheat straw consisted of compact, ordered microfibril bundles with smooth and intact surfaces. After pretreatment, the fibril bundles displayed looser, porous, and partially separated structures. The wheat straw treated with ChCl/BDO/$AlCl_3$ was split into elongated bundles, with fibrils exposed in disordered arrangements. This phenomenon was potentially due to the ternary DES disrupting the lignocellulosic cross-links and partially dissolving or removing the hemicellulose and lignin. These changes increased the cellulose surface area and exposed disorderly elementary fibrils, consistent with the FTIR and XRD analyses. The SEM results confirm that pretreatment caused the wheat straw to

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become more fragmented, enlarged the specific surface area of the fibrils, and increased the accessibility of enzyme binding sites during hydrolysis. Consequently, the enzymatic hydrolysis yield was enhanced (Wu et al., 2024).

4. Conclusions

In this study, ternary DES systems were developed by incorporating solid acids into binary DES frameworks for wheat straw biomass pretreatment. Correlation analysis between the Kamlet-Taft solvatochromic parameters and the lignin extraction rate/enzymatic hydrolysis efficiency revealed that ternary DESs with more substantial hydrogen bond acidity exhibited superior pretreatment performance. Among these DES systems, the ChCl/BDO/$AlCl_3$ (25:50:1) system demonstrated optimal efficacy. Reaction parameter optimization was performed to determine that the best performance was achieved by pretreating the wheat straw at 120 °C for 60 min, which provided an 85.74% lignin extraction rate, 99.67% polysaccharide degradation rate, and 74.59% reducing sugar yield. Moreover, the optimal pretreatment conditions demonstrated broad applicability across diverse biomass types, providing an effective solution for green and efficient lignocellulose separation.

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Table

Table 1. Kamlet-Taft solvatochromatic parameters of prepared DESs

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Composition of DES	Molar ratio of components	π^*	α	β
ChCl/BDO	1/2	1.26±0.05	1.24±0.07	0.49±0.14
ChCl/BDO/$AlCl_3$	25/25/0.5	1.42±0.03	1.71±0.05	0.34±0.03
ChCl/BDO/$AlCl_3$	25/50/0.5	1.35±0.05	2.01±0.08	0.52±0.14
ChCl/BDO/$AlCl_3$	25/50/1	1.53±0.07	2.39±0.09	0.31±0.13
ChCl/EG	1/2	1.14±0.04	1.20±0.07	0.82±0.03
ChCl/EG/PTA	25/25/0.025	1.27±0.11	1.68±0.05	0.62±0.20
ChCl/EG/PTA	25/50/0.025	1.24±0.04	1.93±0.02	0.86±0.12
ChCl/EG/PTA	25/50/0.05	1.39±0.15	2.04±0.13	0.50±0.19
ChCl/GL	1/2	1.17±0.04	1.15±0.05	0.63±0.07
ChCl/GL	1/4	1.09±0.06	1.43±0.08	0.94±0.11
ChCl/GL/$FeBr_3$	25/50/0.25	1.28±0.05	1.46±0.17	0.45±0.03
ChCl/GL/$FeBr_3$	25/100/0.25	1.27±0.09	1.84±0.07	0.53±0.11
ChCl/GL/$FeBr_3$	25/100/0.5	1.44±0.09	1.97±0.07	0.44±0.08

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Figures

Fig. 1. Effects of different DES types and molar ratios on pretreatment and enzymatic hydrolysis of wheat straw.

Fig. 2. Correlations between lignin extraction rate, polysaccharide degradation rate, reducing sugar yield, and Kamlet-Taft solvatochromatic parameters (α and α-β) of different DES pretreatment systems.

Fig. 3. Chemical composition distribution (acid-insoluble lignin (ASL) content, acid-soluble lignin (AIL) content, and polysaccharide content) of lignin (a) and solid residue (b) samples obtained using different DES pretreatment systems.

Fig. 4. Effects of reaction temperature (a) and time (b) on pretreatment and enzymatic hydrolysis.

Fig. 5. Effects of ChCl/BDO/$AlCl_3$ on pretreatment and enzymatic hydrolysis of different biomasses.

Fig. 6. FTIR (a) and XRD (b) analysis of wheat straw before and after different ternary DES pretreatments.

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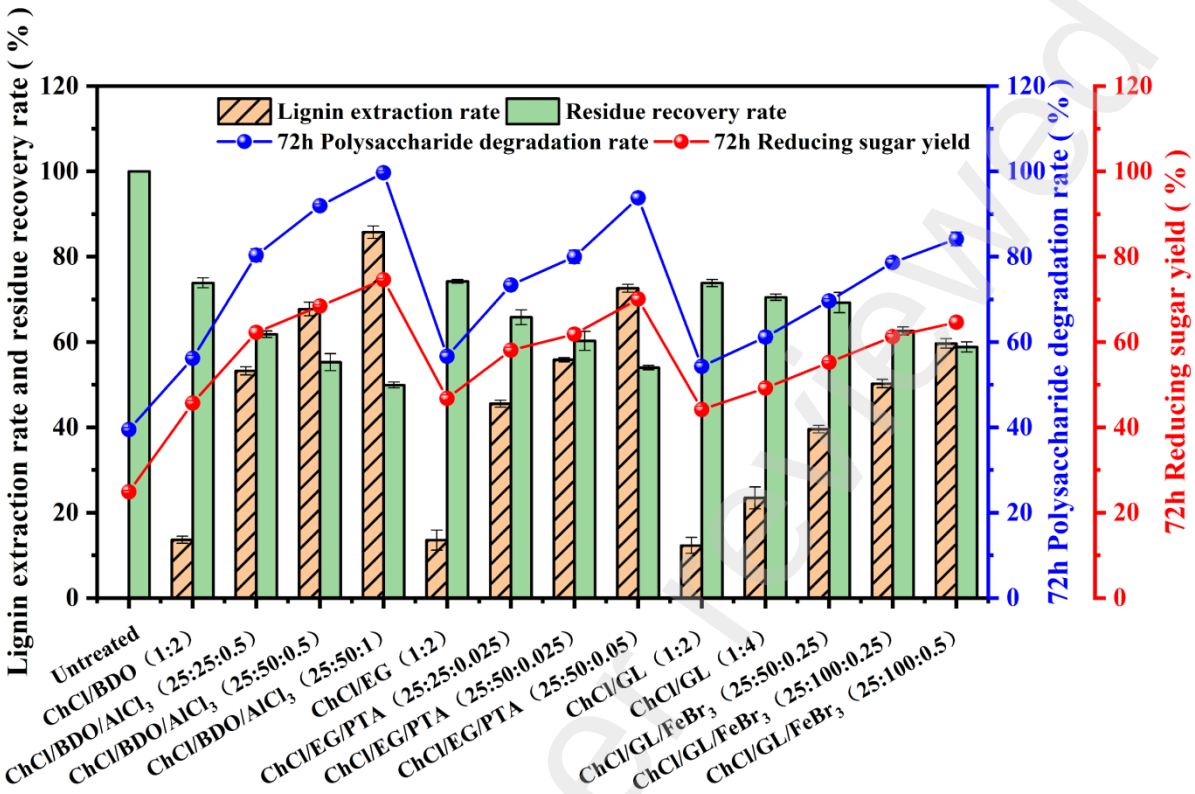


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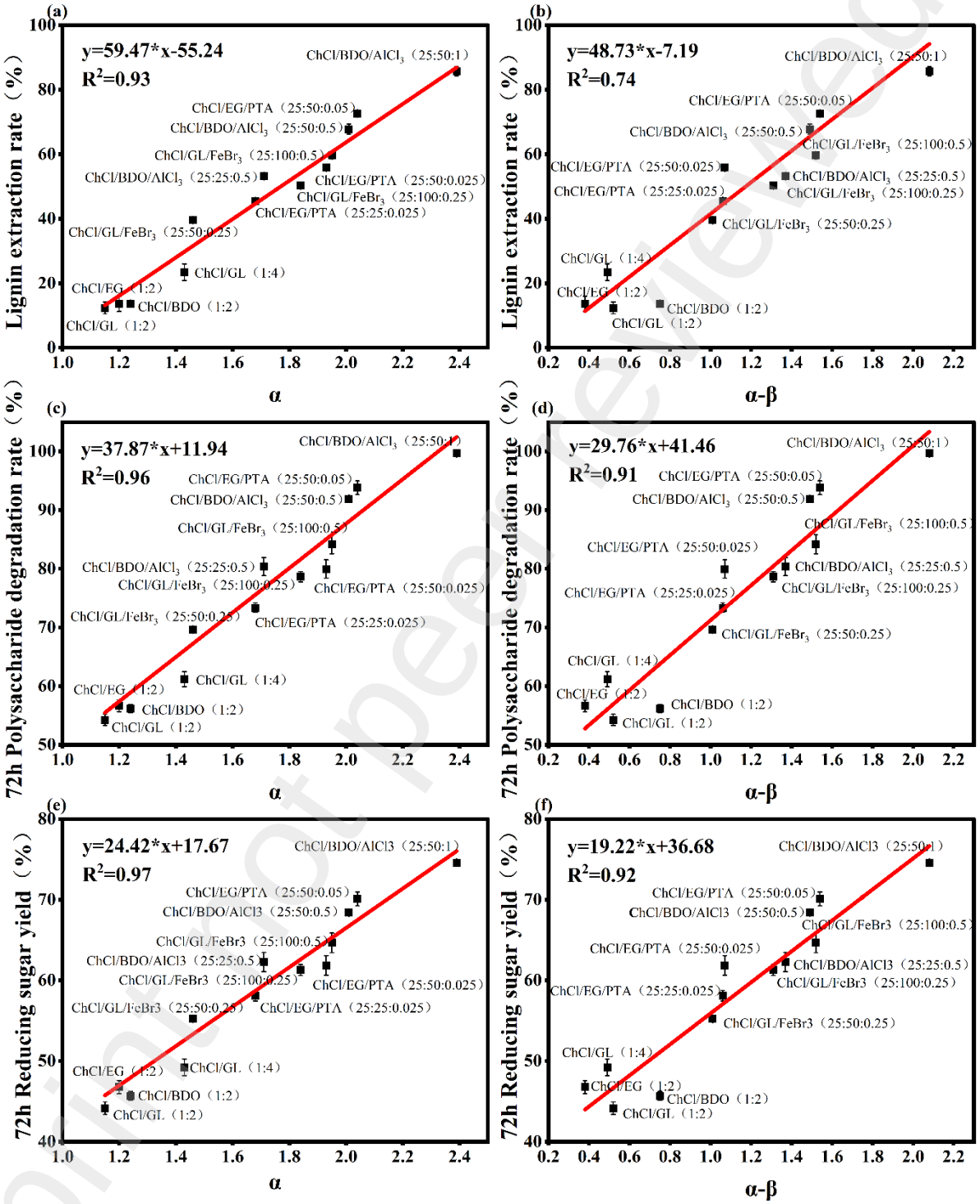


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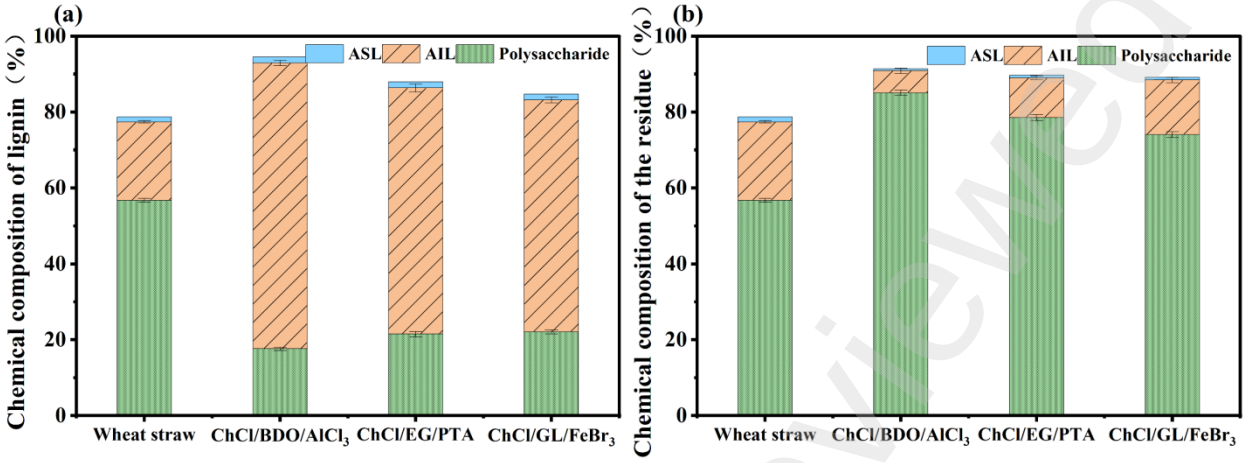


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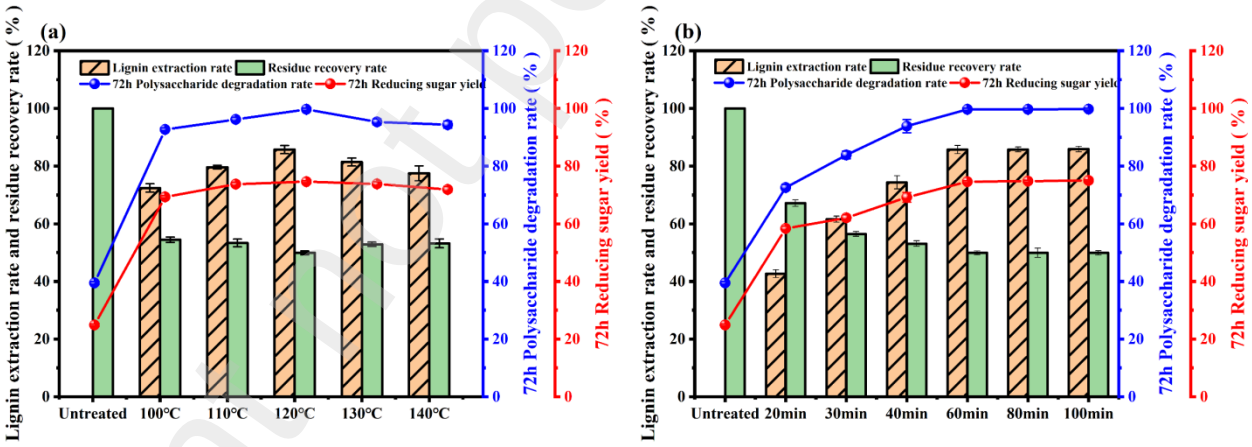


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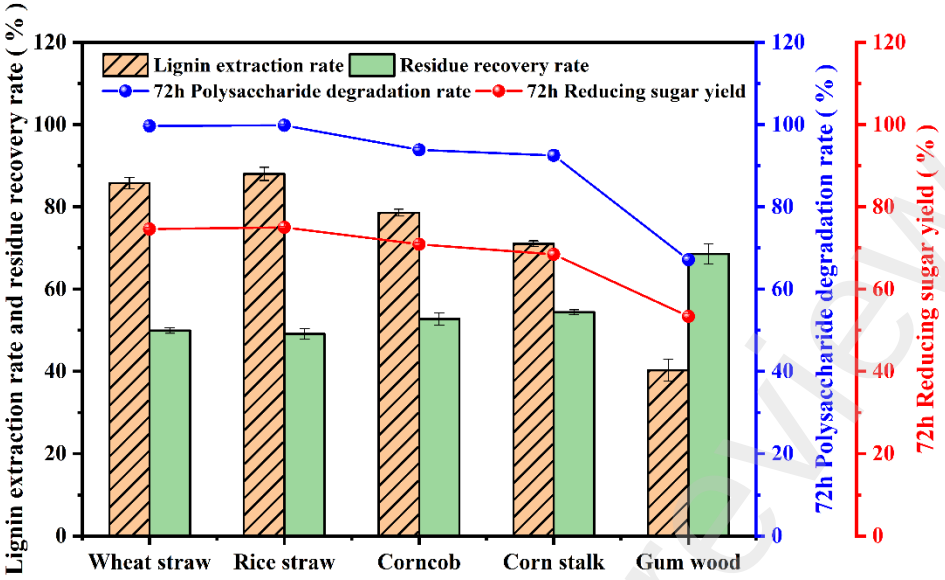


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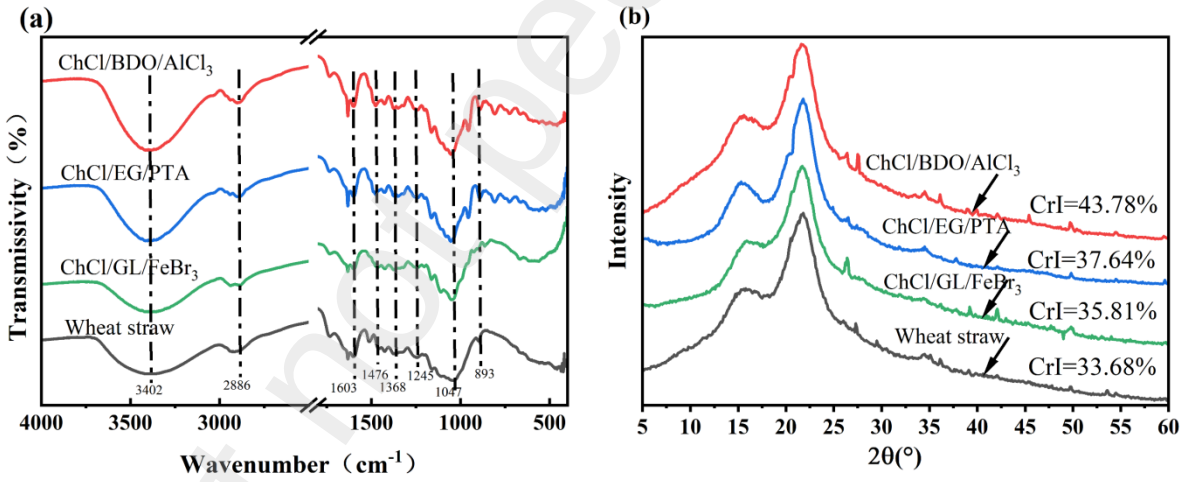


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